ECEN 743: Reinforcement Learning
Assignment 6

Overview

1. You will code this project from scratch. You are free to use any training framework of your choice, but Stable Baselines3 is highly recommended.

2. You have to submit an HTML report, your code, and four videos to Canvas.

3. Put all your files (HTML report, code and videos) into a **single compressed folder** named **LastName_FirstName_A5.zip**.

4. **HPRC Note:** It is highly recommended that you load the Xvfb module by running `ml Xvfb/21.1.14` and use it while performing inference to successfully save your videos (e.g., using `xvfb-run`).

In this assignment, you will implement and evaluate modern Deep Reinforcement Learning algorithms on high-dimensional continuous control tasks. You will use the MuJoCo environments from Gymnasium, specifically focusing on the Pusher and Humanoid environments.

Project Requirements

1. **Environment Setup:** Set up the Gymnasium MuJoCo environments and properly initialize the `Pusher` and `Humanoid` tasks.

2. **Algorithm Training & Hyperparameter Tuning:**

   (a) Train agents using two different algorithms: Proximal Policy Optimization (**PPO**) and Soft Actor-Critic (**SAC**).

   (b) You are responsible for finding the optimal hyperparameters yourself to successfully solve both environments.

3. **Inference & Video Generation:**

   (a) After training, write an inference script that loads your saved models and runs them in the environment.

   (b) You must record and save a total of 4 videos demonstrating your trained policies:
   - PPO solving Pusher
   - SAC solving Pusher
   - PPO solving Humanoid
   - SAC solving Humanoid

4. **HTML Report & Analysis:** Your HTML report must serve as a comprehensive summary of your experiments. It **must** include:

(a) **Hyperparameters:** A clear list or table of the optimal hyperparameters you found and used for both algorithms across both environments.

(b) **Training Graphs:** Plots showing the training performance (e.g., episode reward vs. timesteps) for PPO and SAC.

(c) **Findings & Comparison:** A detailed discussion comparing the performance, sample efficiency, and overall behavior of PPO versus SAC based on your results. Which algorithm performed better on which task, and why?